

Probing the Lack of Stable Internal Beliefs in LLMs

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Abstract

Persona-driven large language models (LLMs) require consistent behavioral tendencies across interactions to simulate human-like personality traits, such as persistence or reliability. However, current LLMs often lack stable internal representations that anchor their responses over extended dialogues. This work explores whether LLMs can maintain "implicit consistency", defined as persistent adherence to an unstated goal in multi-turn interactions. We designed a 20-question-style riddle game paradigm where an LLM is tasked with secretly selecting a target and responding to users' guesses with "yes/no" answers. Through evaluations, we find that LLMs struggle to preserve latent consistency: their implicit "goals" shift across turns unless explicitly provided their selected target in context. These findings highlight critical limitations in the development of persona-driven LLMs and underscore the need for mechanisms that anchor implicit goals over time, which is key to realistic personality modeling in interactive applications, such as dialogue systems.

1 Introduction

Large language models have demonstrated remarkable capabilities in simulating human-like behaviors and personas, enabling their deployment in interactive applications such as dialogue systems, narrative generation, and role-playing assistants. However, a fundamental requirement for realistic persona modeling—consistent behavioral tendencies across extended interactions—remains largely unverified. While existing research has focused on *external consistency* (e.g., factual coherence and logical self-contradiction [1–3]), the ability of LLMs to maintain *internal goal persistence* without explicit reminders is critical for simulating human traits like reliability and determination.

In this work, we investigate whether LLMs possess "implicit consistency"—the capacity to maintain persistent adherence to an unstated goal throughout multi-turn interactions. This form of consistency lies at the core of believable persona modeling: a model that secretly changes its fundamental objectives mid-conversation fails to exhibit genuine personality traits, regardless of surface-level behavioral coherence.

To probe this capability, we design a 20-question-style dialogue game [4] where an LLM secretly selects a target entity and must answer yes/no questions while maintaining consistency with its initial choice. This paradigm creates a controlled setting to distinguish between *external inconsistency* (observable self-contradictions) and *implicit inconsistency* (internal goal drift despite externally coherent responses). Our experiments reveal a striking finding: LLMs universally struggle with implicit consistency, with most models exhibiting goal drift in 100% of dialogues and changing their internal targets in 16.77% to over 50% of conversation turns. This fundamental limitation

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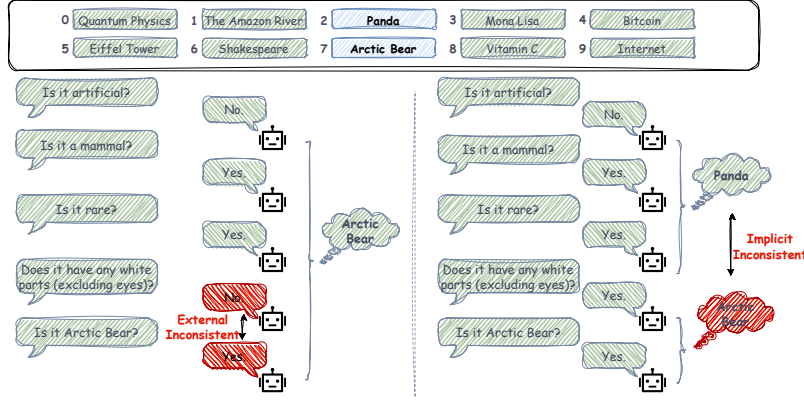


Figure 1: An example of external & implicit inconsistencies. See Appendix A for explanation.

persists across both simple number guessing and semantically rich entity guessing tasks, and is often exacerbated by reasoning-enhanced model variants.

The main contributions of this work are: (1) We introduce and formalize the concept of *implicit consistency*, distinguishing it from external consistency through mathematical definitions and a diagnostic framework. (2) We develop a unified probing methodology that enables continuous monitoring of internal belief states across diverse tasks, revealing pervasive goal drift in state-of-the-art LLMs. (3) We demonstrate that explicit training with KL divergence regularization reduces implicit inconsistency, providing a possible pathway toward more consistent persona-driven models.

Our findings suggest that building truly consistent persona-driven LLMs requires explicit mechanisms for anchoring implicit goals, moving beyond surface-level behavioral alignment to address fundamental limitations in internal state stability.

2 Taxonomy and Formalisms for Inconsistency in LLMs

We study LLM’s consistency through a 20-question-style [4] dialogue game (see Figure 1), where the model secretly selects a target and answers yes/no questions to help the user guess it. This paradigm allows us to observe different types of inconsistencies that may arise during interactions.

Task Definition. Let M be a LLM, and $E = \{e_1, e_2, \dots, e_n\}$ denote a set of candidate entities. At the start of a multi-turn dialogue, the model M is instructed to secretly select a target entity $e_t \in E$ and maintain this choice consistently throughout the interaction. A dialogue consists of a sequence of T question-answer pairs: $D = \{(q_1, a_1), (q_2, a_2), \dots, (q_T, a_T)\}$. The model’s *implicit goal* at turn i is represented by its internal belief state $b_i \in \Delta(E)$, where $\Delta(E)$ denotes the space of possible distributions over the candidate set E .

External Inconsistency. Occurs when the model’s responses contradict its *own previous statements* within the same dialogue, regardless of ground truth. Formally, there exist turns i and j such that: $a_i \wedge a_j \models \perp$, where $\perp$ denotes logical contradiction, and the contradiction arises from the semantic conflict between questions q_i and q_j that reference the same underlying property.

Implicit Inconsistency. Occurs when the model’s *internal belief state* drifts from the original target e_t to a different entity $e' \in E$ during the dialogue, while maintaining external response consistency. Formally, there exists a turn k such that: $\arg \max_{e \in E} b_k = e' \neq e_t$, where $b_k \in \Delta(E)$ represents the model’s belief distribution over candidate entities at turn k . The drift from e_t to e' remains hidden until explicit revelation, as for all turns $i < d$ (where d is the guess turn), the answers satisfy: $\forall i < d, f(q_i, e_t) = f(q_i, e')$, where $f(q, e)$ gives the real answer to question q about entity e .

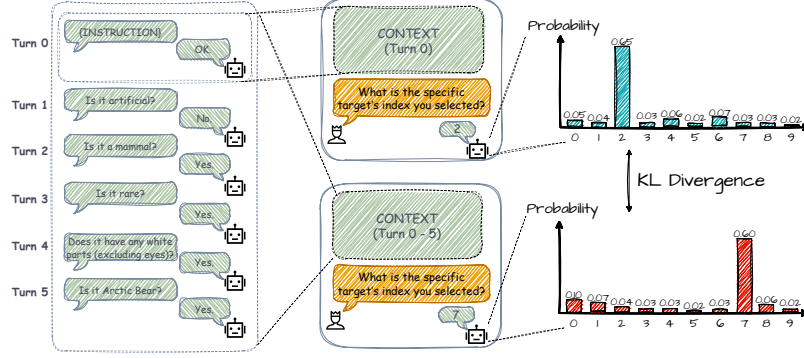


Figure 2: An example showing target index probings and KL divergence calculations across turns.

3 Proposed Method

3.1 Task Category

We design two variants of the 20-question-style game to evaluate LLMs’ implicit consistencies. In both tasks, we employ two model roles: the **Proposer** that secretly selects and maintains a target, and the **Guesser** that asks yes/no questions to identify the target. Both tasks employ model-model dialogues to ensure controlled experimental conditions and scalable evaluation across multiple LLMs.

Number Guessing. The Proposer is provided with 10 numbers randomly sampled from 0 to 99 and secretly selects one as its target. The Guesser then attempts to identify the target through yes/no questions. This task provides a controlled setting with minimal semantic complexity, allowing us to isolate the consistency phenomenon. We explore both prompt-based evaluation and supervised fine-tuning for this task.

Entity Guessing. The Proposer is provided with 10 candidate entities spanning diverse categories and secretly selects one as its target. The Guesser attempts to identify the entity through yes/no questions. This task evaluates consistency in semantically rich scenarios that more closely resemble real-world applications, using prompt-based evaluation only.

3.2 Probing Technique

To enable precise measurement of belief states across diverse tasks, we develop a unified probing framework based on numerical indexing. Each candidate entity is mapped to a single-token index (0-9), which avoids the complications of analyzing probabilities across multiple tokens when using full entity names. This indexing scheme creates a consistent probabilistic representation that allows us to compare belief dynamics across different task domains and model architectures.

Building on this foundation, we implement a branch-based probing mechanism (See Figure 2). After each main dialogue turn, we create an isolated context containing the full dialogue history and pose the probe: “What is the specific target’s index you selected?” This design enables continuous monitoring of the Proposer’s internal belief state $b_i \in \Delta(E)$ without contaminating the main interaction, providing clean measurements for our consistency analysis.

3.3 Metric Definition

We quantify implicit consistency via three complementary aspects to capture different facets of goal stability, with additional supplementary metrics detailed in the Appendix C.

Token-based. Tracks discrete changes in the model’s implicit target selection. The key metric is **Drift Rate** ($\frac{\# \text{ changes }}{T}$), which calculates how frequently the target changes throughout the dialogue.

Probability-based. Analyzes continuous shifts in the model’s implicit target distribution. We use **KL Divergence** ($D_{\text{KL}}(P_t \parallel P_{t-1})$) to quantify turn-wise changes in the model’s belief state.

External Consistency. Evaluates surface-level coherence using **External Consistency Verification** ($\mathbb{E}[\mathbb{I}(\text{no violation})]$), a LLM-as-a-Judge approach following [1].

3.4 Training Framework

For the number guessing task, we employ supervised fine-tuning using data from our prompt-based experiments. We curate the training set by selecting dialogues that exhibit no external inconsistencies, ensuring that the models we train at least maintain surface-level coherence. The training objective combines KL divergence and cross-entropy losses: $\mathcal{L} = \sum_{t=1}^T [\alpha \cdot D_{\text{KL}}(P_{\text{probe}}^t \| P_{\text{initial}}) + \beta \cdot \mathcal{L}_{\text{CE}}]$, where P_{initial} is the belief distribution from the first probe, P_{probe}^t is the distribution at turn t , and $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss for the probing responses. We also experiment with each loss component in isolation ($\alpha = 0$ or $\beta = 0$) to analyze their individual effects on consistency improvement.

4 Experimental Results

4.1 Prompting

Table 1: Prompting results. "+ Reasoning" stands for using the reasoning model as the Proposer.

Model	Number Guessing				Entity Guessing			
	D.R. (↓)	Once D.R. (↓)	KL Div. (↓)	E.C.V. (↓)	D.R. (↓)	Once D.R. (↓)	KL Div. (↓)	E.C.V. (↓)
GPT-4o	43.36 $\pm$ 11.72	100.00 $\pm$ 0.00	0.66 $\pm$ 0.27	0.00 $\pm$ 0.00	22.64 $\pm$ 8.46	100.00 $\pm$ 0.00	0.75 $\pm$ 0.43	0.00 $\pm$ 0.00
Seed-1.6	17.37 $\pm$ 5.62	95.24 $\pm$ 21.30	0.99 $\pm$ 0.38	42.86 $\pm$ 49.49	11.31 $\pm$ 4.52	100.00 $\pm$ 0.00	0.17 $\pm$ 0.05	0.00 $\pm$ 0.00
+ Reasoning	28.57 $\pm$ 7.49	94.12 $\pm$ 23.53	-	0.00 $\pm$ 0.00	11.05 $\pm$ 5.18	95.12 $\pm$ 21.54	-	4.88 $\pm$ 21.54
Deepseek-v3.1	38.46 $\pm$ 13.71	100.00 $\pm$ 0.00	0.62 $\pm$ 0.23	80.95 $\pm$ 39.27	37.68 $\pm$ 11.38	100.00 $\pm$ 0.00	0.09 $\pm$ 0.04	41.46 $\pm$ 49.27
+ Reasoning	54.25 $\pm$ 21.56	95.24 $\pm$ 21.30	-	28.57 $\pm$ 45.18	27.01 $\pm$ 13.54	100.00 $\pm$ 0.00	-	7.32 $\pm$ 26.04
Claude-3.7-Sonnet	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	-	0.00 $\pm$ 0.00	12.25 $\pm$ 2.11	100.00 $\pm$ 0.00	-	2.86 $\pm$ 16.66
+ Reasoning	71.15 $\pm$ 18.30	100.00 $\pm$ 0.00	-	0.00 $\pm$ 0.00	33.42 $\pm$ 12.61	100.00 $\pm$ 0.00	-	2.44 $\pm$ 15.43

Our experiments reveal pervasive implicit inconsistency across all tested LLMs (Table 1; see Appendix D for details). Most models exhibit 100% Once Drift Rate (See Appendix C), with Drift Rates ranging from 17.37% to 100%. We hypothesize that reasoning-enhanced models increase drift in simple tasks (like Number Guessing) due to overthinking—introducing unnecessary complexity in straightforward scenarios. Claude’s reversed pattern may suggest architecture-specific differences in how reasoning modules interact with goal maintenance; we keep it as our future work. Notably, reasoning improves external consistency in some cases (e.g., Deepseek-v3.1 reduces violations), indicating that while reasoning may destabilize implicit goals, it can enhance surface-level coherence.

4.2 Supervised Fine Tuning

Due to computational constraints, we fine-tune Qwen-2.5-14B-Instruct as Proposer with Seed-1.6-Reasoning as Guesser. Results (See Table 2) show CE-only fine-tuning fails to mitigate implicit inconsistency, while KL-only significantly improves goal persistence. The combined CE+KL approach yields the compromise results, indicating KL divergence effectively regularizes belief stability.

Table 2: Qwen-2.5-14B-Instruct variants metric results in Number Guessing.

Model	D.R. (↓)	O.D.R. (↓)	KL Div. (↓)	E.C.V. (↓)
Qwen-2.5-14B-Instruct	36.83 $\pm$ 10.72	100.00 $\pm$ 0.00	3.30 $\pm$ 3.15	100.00 $\pm$ 0.00
+ CE	31.63 $\pm$ 18.78	76.19 $\pm$ 42.59	0.47 $\pm$ 0.26	14.29 $\pm$ 34.99
+ KL	13.99 $\pm$ 9.04	95.24 $\pm$ 21.30	1.41 $\pm$ 1.11	100.00 $\pm$ 0.00
+ CE & KL	14.48 $\pm$ 5.08	95.24 $\pm$ 21.30	1.46 $\pm$ 0.97	95.24 $\pm$ 21.30

5 Conclusion and Future Work

In this work, we identified and formalized a critical limitation in current LLMs: the inability to maintain implicit consistency—stable adherence to unstated goals across multi-turn interactions. Through a carefully designed 20-question paradigm and unified probing framework, we demonstrated